Opera File Edit View History Bookmarks Developer Window Help zoom 66% Sat 9 Mar 2:23 PM
You are screen sharing Stop Share

Stemming — reduces the Vocab

Nouns are not cut-
Verbs, adjectives ..

running, runs, ran → run
1 2 3 1

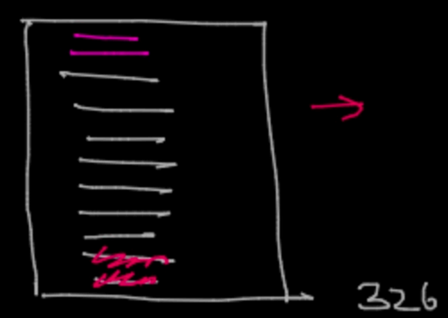
plural ⇒ Singular

Vocab → unique words in the Corpus

Btw, actually, basically, ...

Search → How to create a linear dis.

Linear review example → stop words



Opera File Edit View History Bookmarks Developer Window Help zoom 87% Sat 9 Mar 5:11 PM
You are screen sharing Stop Share

Recommendations

Association Rule

Recommendn Systems

(1) Camera → similar → Recommendn system

(2) ppl who bought this also bought →

Association Rule Mining.

Content Based ⇓ Regression Model. Classification Mod.

Collaborative filtering

Similarity Based algorithms

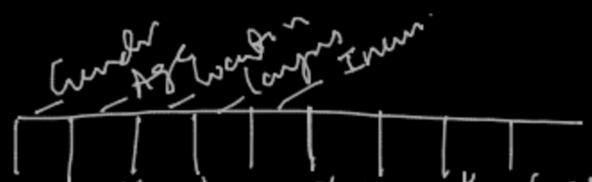
User-User Similarity

Item-Item Similarity

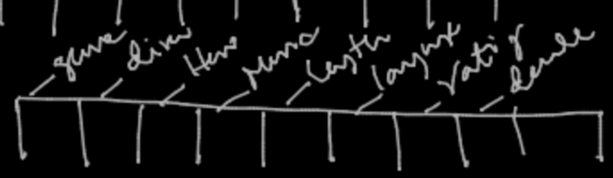
Netflix → Movie Recommendations

Users → Customer } Features
Items → Movies

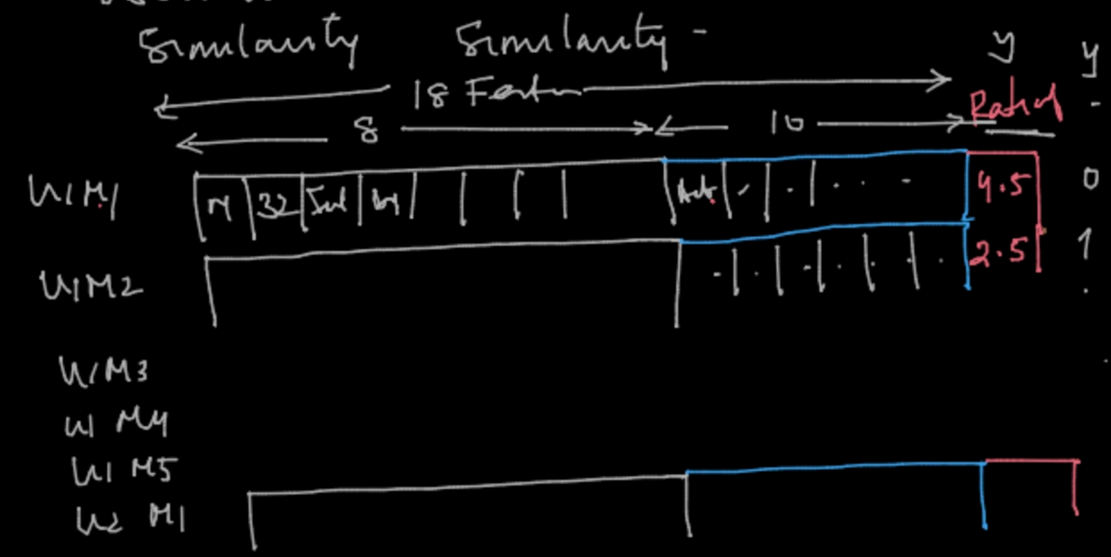
User =



Movie =



10 users
5 Movies



Opera File Edit View History Bookmarks Developer Window Help
You are screen sharing Stop Share
zoom 100% Sat 9 Mar 3:18 PM

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	y ⇒ Rate Watch
1	M	32	Ind	Eng	Action	div	2hr	Eng	4.5	1
2	M	32	In	tay	Rom	div	132	HIN	2.5	0
.										1
.										1
										0
										0
n										6

⇒ Regression Model

10 users ✓
5 Movies ✓
→ 6th Movie

X_test

									y	
u1	M6	.							4.0 ✓	1 ✓
u2	M6	.							3.0 ✗	0 ✗
u3	M6	-	-	-	-	-	-	-	4.5 ✓	1 ✓
u4	M6								2.5 ✗	0
									⋮	1
u10	M6									6

zoom.us Meeting View Edit Window Help
You are screen sharing Stop Share
Sat 9 Mar 3:53 PM

Collaborative Filtering → (user-user similarity)

$$\begin{array}{c|c} & \begin{matrix} I_1 & I_2 & I_3 & \cdots & I_m \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \vdots \\ u_n \end{matrix} & \begin{bmatrix} R_{11} & R_{12} & R_{13} & \cdots & R_{1m} \\ R_{21} & R_{22} & R_{23} & \cdots & R_{2m} \\ & & & 4.5 & \\ R_{41} & R_{42} & R_{43} & \cdots & R_{4m} \\ & & & & \\ R_{n1} & R_{n2} & R_{n3} & \cdots & R_{nm} \end{bmatrix} \end{array}$$

$n \times m$

5M 2M

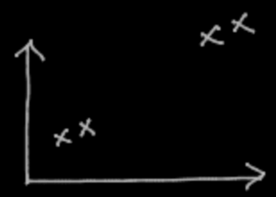
→ Preferance of u_4

How customers like this item

⇒ Sparse Matrix.

$$\begin{array}{c|c} & \begin{matrix} M_1 & M_2 & M_3 & \cdots & M_{2M} \end{matrix} \\ \begin{matrix} u_1 \\ \\ \\ \\ u_{5M} \end{matrix} & \begin{bmatrix} 0 & 0 & 4.5 & 0 & 0 & \cdots & 3.0 & \cdots \\ \cdots & & & & & & & \\ & & & & 3.0 & & & \\ & & & & & & & \\ & & & & & & & \end{bmatrix} \end{array}$$

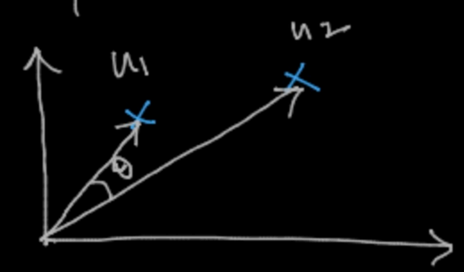
How similar are u_1 & u_2

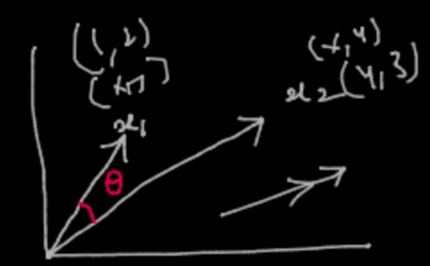


⇒ Distance $\propto \frac{1}{\text{Similarity}}$

Cosine Similarity

⇒ Matrix Completion problem



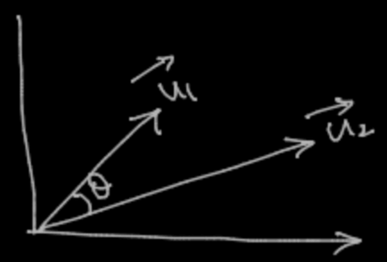


$\cos(\theta) = 1$

$x_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $x_2 = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$

PowerPoint File Edit View Insert Format Arrange Tools Slide Show Window Help zoom 78% Sat 9 Mar 5:35 PM
You are screen sharing Stop Share

$$\vec{u}_1 = \begin{bmatrix} R_{11} \\ R_{12} \\ R_{13} \\ R_{14} \\ \vdots \\ R_{1m} \end{bmatrix} \qquad \vec{u}_4 = \begin{bmatrix} R_{41} \\ R_{42} \\ R_{43} \\ \vdots \\ R_{4m} \end{bmatrix}$$



$\cos\theta \rightarrow$ Cosine Similarity betn u_1 & u_4

$0 \longrightarrow 1$

$0.1, 0.2, \ldots 0.7, 0.8$

$n = 5$

usu - usu Similarity Matrix

$$\begin{array}{c|c} & u_1 \quad u_2 \quad u_3 \quad \cdots \quad u_n \\ \hline \begin{array}{c} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_n \end{array} & \begin{bmatrix} 1.0 & & & & \\ S_{21} & 1.0 & & & \\ S_{31} & S_{31} & 1.0 & & \\ \vdots & \vdots & \vdots & & \\ S_{n1} & S_{n2} & S_{n3} & \cdots & S_{n1} \quad 1.0 \end{bmatrix} \end{array}$$

usu - m

$$\begin{array}{c|c} & u_1 \quad u_2 \quad u_3 \quad u_4 \quad u_5 \\ \hline \begin{array}{c} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{array} & \begin{bmatrix} 0 & S_{12} & S_{13} & S_{14} & S_{15} \\ 0.2 & 0 & 0.6 & 0.3 & 0.9 \\ S_{31} & S_{32} & 0 & S_{34} & S_{35} \\ S_{41} & S_{42} & S_{43} & 0 & S_{45} \\ S_{51} & S_{52} & S_{45} & S_{54} & 0 \end{bmatrix} \end{array} \quad 5 \times 5$$

$u_4 \rightarrow$ what movies?
Find ppl similar to u_4
which movies they liked

20

(1) Which user is most similar to u_2
Which movies should we recommend for u_2?

$u_2 \Rightarrow u_5$
Which movie did u_5 recommend highly

$\boxed{M_7, M_{23}} \rightarrow 4.5$

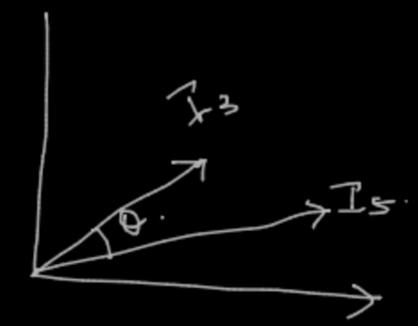
5 / 10

zoom.us Meeting View Edit Window Help
You are screen sharing Stop Share
Sat 9 Mar 3:54 PM

$$\begin{array}{c|c} & I_1\ I_2\ I_3\ \cdots\ I_m \\ \hline \begin{array}{c} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_n \end{array} & \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \end{array}$$

Item Item Similarity -

$$\vec{I}_3 = \begin{bmatrix} R_{13} \\ R_{23} \\ R_{33} \\ \vdots \\ R_{n3} \end{bmatrix} \qquad I_5 \begin{bmatrix} R_{15} \\ R_{25} \\ R_{35} \\ \vdots \\ R_{n5} \end{bmatrix}$$



$\cos\theta \Rightarrow$

$$\begin{array}{c|c} & I_1\ I_2\ I_3\ \cdots\ I_m \\ \hline \begin{array}{c} I_1 \\ I_2 \\ I_3 \\ \vdots \\ I_m \end{array} & \begin{bmatrix} 1.0 & 0.5 & 0.9 & \cdots & 0.6 \\ 0.5 & 1.0 & & & \\ 0.3 & & 1.0 & & \\ \vdots & & & & \\ 0.9 & & & & 1.0 \end{bmatrix} \end{array}$$

$I_1 \Rightarrow I_6, I_7$

↳ Find users who rated I_1 highly

Recommend I_6 & I_7

zoom.us Meeting View Edit Window Help
zoom 100% Sat 9 Mar 4:03 PM
You are screen sharing Stop Share

Approximations → local movies / users

u_1 u_2 . . . v_3

1. When to recommend

— real time msecs → approx but fast

— delayed → Accurate

10, u_{11} →

Cold start → popular movies

→ Randomly Rem

→ Influencers / ...

?

4.20 pm

Opera File Edit View History Bookmarks Developer Window Help zoom 92% Sat 9 Mar 4:59 PM
You are screen sharing Stop Share

Coding.

$$\text{user_movie_matrix} = \begin{array}{c} \\ u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_{408} \end{array} \begin{array}{c} \begin{array}{cccc} M_1 & M_2 & M_3 \cdots & M_{10} \end{array} \\ \begin{bmatrix} \text{Ratings} \\ \\ R_{31} \; R_{32} \; R_{33} \cdots R_{3m} \\ \\ \\ \end{bmatrix} \end{array}_{408 \times 10}$$

$$\text{user_sim_matrix} = \begin{array}{c} \\ u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_{408} \end{array} \begin{array}{c} \begin{array}{cccc} u_1 & u_2 & u_3 \cdots & u_{408} \end{array} \\ \begin{bmatrix} 1.0 & \cdots & & \\ \cdots & 1.0 & & \\ & & 1.0 & \\ & & & \ddots \\ & & & & 1.0 \end{bmatrix} \end{array}$$

np.fill_diagonal(, 0)

Find 3 users who are very similar to u_1 →

PowerPoint File Edit View Insert Format Arrange Tools Slide Show Window Help
You are screen sharing Stop Share
zoom 78% Sat 9 Mar 5:34 PM

Association Rule Mining -

— Market Basket Analysis -

Find meaningful associations

If Antecedent ⇒ then Consequent

If Bread ⇒ then Butter
(A) (C)

$$\text{Support} = \frac{\#(A, C)}{\text{Total No. of transaction}} = \frac{80}{100} \Rightarrow 80\%$$

$$\text{Confidence} = \frac{\#(A, C)}{(\# A)} = \left(\frac{80}{90}\right)$$

A

90 → Bread

80 ⇒ Bread + Butter
A C

Apriori Algorithm ⟹ Apriori principle

9 / 10

PowerPoint File Edit View Insert Format Arrange Tools Slide Show Window Help zoom 78% Sat 9 Mar 5:33 PM
You are screen sharing Stop Share

Apriori Principle

"If an itemset is frequent all its subsets should also be frequent"

Itemset = {Butter} → 1 item itemset

{Bread, Butter} ⇒ 2 item itemset

Frequent Item Set ⇒ Item Set for which Support > threshold-value

10 / 10